CI/CD HANDS-ON LAB | BEGINNER

Jenkins First Job Creation

Freestyle Project: build a job from the Jenkins UI, no Jenkinsfile needed

~15 min | No code needed | Linux shell

In this lab you will configure a simple Jenkins job that prints messages to the console, run it manually, and read the result. It is the quickest way to see how Jenkins jobs, build steps and logs fit together.

1 The workflow at a glance

2 Step-by-step implementation

STEP 1 Open Jenkins

- Start your Jenkins server.
- Open the Jenkins URL in your browser, for example `http://localhost:8080`
- Log in with your Jenkins credentials.
- You should see the **Jenkins Dashboard**.

STEP 2 Create a new job

- Click **New Item** in the left-side menu.
- Enter the job name: `CloudMessenger-First-Job`
- Select **Freestyle project**.
- Click **OK**.

> **What is a Freestyle project?**
> A simple Jenkins job you configure through the graphical interface by selecting build steps and options.

STEP 3 Configure the job

You will see several configuration sections:

General	Add a description of your job.
Source Code Management	Leave **None** for this first lab.
Build Triggers	Leave unchecked for now.
Build Environment	Keep the default settings.
Build Steps	Where you tell Jenkins what to execute.
Post-build Actions	Leave empty for now.

In the **Description** field, enter:

```
My first Jenkins Freestyle project to understand job execution.
```

STEP 4 Add a build step

- Scroll to **Build Steps** and click **Add build step**.
- Select **Execute shell** if Jenkins is running on Linux.
- Enter this script:

```
#!/bin/bash

echo "================================="
echo "Welcome to Jenkins!"
echo "Project: Cloud Messenger"
echo "This is my first Freestyle Job."
echo "Learning CI/CD step by step!"
echo "================================="

echo "Current user:"
whoami

echo "Current date:"
date

echo "Current working directory:"
pwd

echo "Build completed successfully!"
```

Important
If Jenkins runs on Windows, use **Execute Windows batch command** instead and enter Windows-compatible commands. Execute Shell is intended for Linux/Unix agents.

STEP 5 Save the job

- Scroll to the bottom and click **Save**.
- Jenkins opens your job's page.
- Your job is configured, but it has **not run yet**.

STEP 6 Execute the job

- Click **Build Now** in the left-side menu.
- Look for a new build number such as **#1** under Build History.
- Click the build number, then click **Console Output**.
- You should see output similar to:

```
Started by user
Building in workspace ...
Welcome to Jenkins!
Project: Cloud Messenger
This is my first Freestyle Job.
Learning CI/CD step by step!
Current user:
jenkins
Current date:
...
Build completed successfully!
Finished: SUCCESS
```

The exact username, directory, date and log lines may differ depending on your setup.

3 What just happened

Term	Simple meaning
Job / Project	A task Jenkins is configured to execute
Freestyle project	A job configured mainly through the UI
Build step	A command or action Jenkins runs
Build Now	Manually starts a build
Build History	Shows previous job executions
Console Output	Displays the execution logs
SUCCESS	Jenkins reports that the build completed successfully

4 Troubleshooting common errors

Execute shell is unavailable	Check the job configuration and agent type; use the build step that fits your operating system.
Permission denied	Check the Jenkins agent user's permissions.
Build marked FAILURE	Open Console Output and inspect the first relevant error.
Job stuck in the queue	Check whether a suitable Jenkins executor or agent is available.

5 Your first-job checklist

- [ ] Opened the Jenkins dashboard
- [ ] Created a Freestyle project
- [ ] Added an Execute Shell build step
- [ ] Saved the configuration
- [ ] Clicked Build Now
- [ ] Viewed Console Output and confirmed SUCCESS

What to learn next

Create a second Freestyle job that checks out source code from GitHub, runs a shell script, and archives a build artifact. This introduces the next steps toward a basic CI pipeline.